

WASP-43b

R-2116

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150413 · created 2026-08-11T00:43:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457125.71971 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.045
Transit depth (Rp/Rs)^2	0.47 +/- 0.62 [%]
Orbital Inclination (inc)	86.98 +/- 2.92
Airmass coefficient 1 (a1)	0.927 +/- 0.023
Airmass coefficient 2 (a2)	0.052 +/- 0.025
Scatter in the residuals of the lightcurve fit is	2.52 %
Best Comparison Star	#1 - [43, 45]
Optimal Aperture	3.1
Optimal Annulus	9.61
Transit Duration (day)	0.054 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.068 $\pm$ 0.045 vs 0.1594 (deviation 1.61 σ)
Duration fit vs published	0.054 d vs 0.0512 d (deviation 0.47 σ)
Mid-time O-C	2.0 min
Depth SNR	1.2
Residual scatter	2.52 %

Light curve

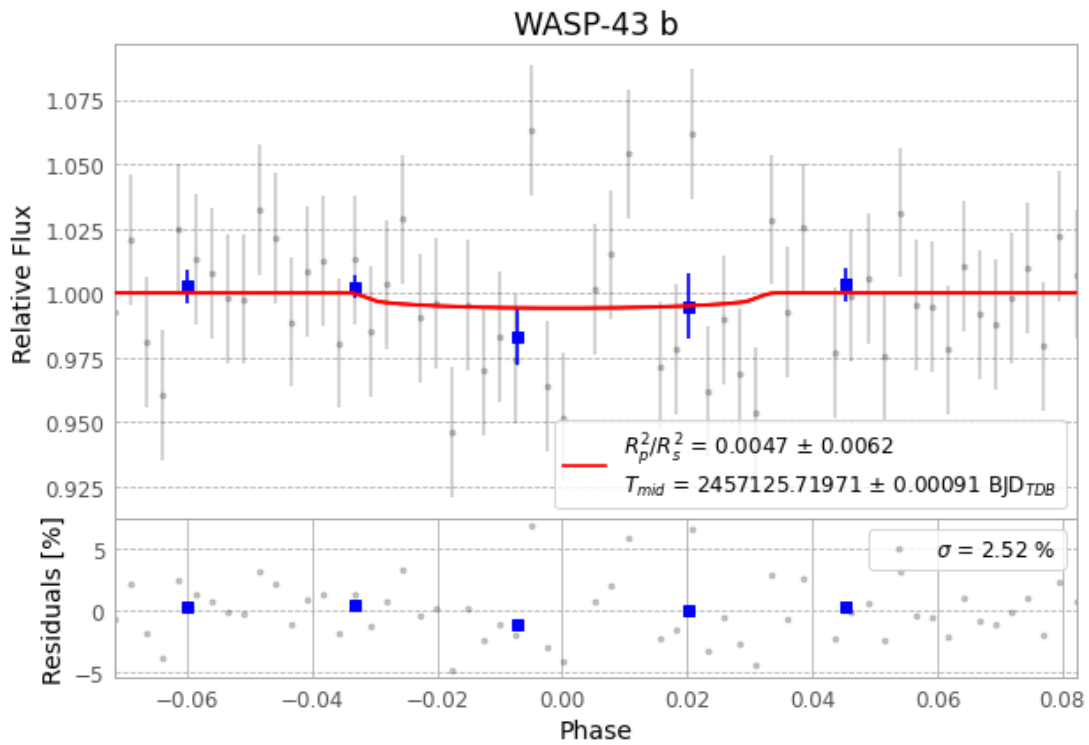


Figure 1 — FinalLightCurve_WASP-43 b_2015-04-12.png (SHA-256 426328e106771031...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [120, 237], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 96873169a4923aeb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), WASP-43150413034512.FITS → WASP-43150413064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150413.log (008023439779...), FinalLightCurve_WASP-43 b_2015-04-12.png (426328e10677...), FinalLightCurve_WASP-43 b_2015-04-12.csv (e7d6e5fe0c3b...)

Compute resources

Wall time 163.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:43Z.